Yarim va chorak doira perimetri

Yarim doira shaklidagi oyna chetiga qancha rezina qistirma kerak? Buning uchun oynaning perimetrini topamiz.

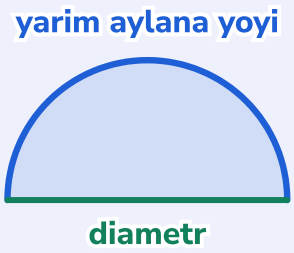
1-QISM

Mavzu

1-QOIDA · YARIM DOIRA PERIMETRI

Shaklni o'rab turgan chiziq uzunligi — uning **perimetri**.

Yarim aylana yoyi — doira aylanasining yarmiga teng bo'lgan egri chiziq.



$$\text{Yarim aylana yoyi uzunligi} = \frac{1}{2} \cdot \pi d$$

$$\text{Yarim aylana perimetri} = \\ = \text{yarim aylana yoyi uzunligi} + \text{diametr}$$

1-MISOL · DIAMETR BERILGAN

Yarim doiraning diametri 12 cm. Uning perimetrini toping ($\pi = 3{,}14$).



1. Yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 12 = 18{,}84$ cm.
2. Diametrni qo'shamiz: $18{,}84 + 12 = 30{,}84$ cm.

2-MISOL · RADIUS BERILGAN

Yarim doiraning radiusi 10,5 m. Perimetrini toping ($\pi = \frac{22}{7}$).



1. Diametr: $2 \cdot 10{,}5 = 21$ m.
2. Yoy: $\frac{1}{2} \cdot \frac{22}{7} \cdot 21 = 33$ m.
3. Perimetr: $33 + 21 = 54$ m.

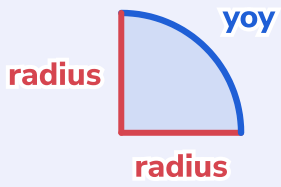
Yarim doiraning perimetri aylana uzunligining yarmi emas. Yoyga diametr ham qo'shiladi.



2-QOIDA · CHORAK DOIRA PERIMETRI

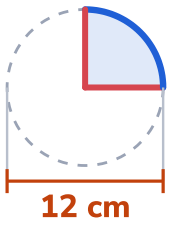
$$\text{Yoy uzunligi} = \frac{1}{4} \cdot \pi d = \frac{1}{4} \cdot 2\pi r$$

$$\text{Chorak aylananing perimetri} = \text{yoy uzunligi} + \text{radius} + \text{radius}$$



1-MISOL · DIAMETR BERILGAN

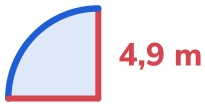
Doiraning diametri 12 cm. Uning choragi perimetrini toping ($\pi = 3{,}14$).



1. Yoy: $\frac{1}{4} \cdot 3{,}14 \cdot 12 = 9{,}42$ cm.
2. Radius: $12 : 2 = 6$ cm.
3. Perimetr: $9{,}42 + 6 + 6 = 21{,}42$ cm.

2-MISOL · RADIUS BERILGAN

Chorak doiraning radiusi 4,9 m. Uning perimetrini toping ($\pi = \frac{22}{7}$).



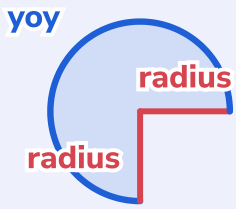
1. Yoy: $\frac{1}{4} \cdot 2 \cdot \frac{22}{7} \cdot 4{,}9$. Avval $4{,}9 : 7 = 0{,}7$. So'ng $\frac{1}{4} \cdot 2 \cdot 22 \cdot 0{,}7 = 7{,}7$ m.
2. Perimetr: $7{,}7 + 4{,}9 + 4{,}9 = 17{,}5$ m.



Yarim doirada to'g'ri chiziq bitta — diametr. Chorak doirada esa ikkita — radius va radius.

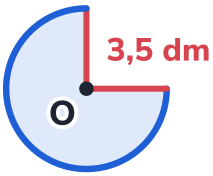
USUL · AYLANANING $\frac{3}{4}$ QISMI

1. $\text{Yoy uzunligi} = \frac{3}{4} \cdot 2\pi r.$
2. $\text{Shaklning perimetri} = \\ = \text{yoy uzunligi} + \text{radius} + \text{radius}.$



1-MISOL · AYLANANING $\frac{3}{4}$ QISMI

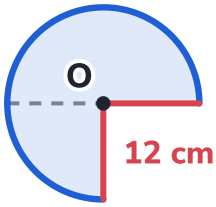
Shakl — aylananing $\frac{3}{4}$ qismi, radiusi 3,5 dm. Perimetrini toping ($\pi = \frac{22}{7}$).



1. Yoy: $\frac{3}{4} \cdot 2 \cdot \frac{22}{7} \cdot 3{,}5$. Avval $3{,}5 : 7 = 0{,}5$. So'ng $\frac{3}{4} \cdot 2 \cdot 22 \cdot 0{,}5 = 16{,}5$ dm.
2. Perimetr: $16{,}5 + 3{,}5 + 3{,}5 = 23{,}5$ dm.

2-MISOL · YARIM VA CHORAK DOIRA

Shakl yarim doira va chorak doiradan iborat, markazi O, radiusi 12 cm. Perimetrini toping ($\pi = 3{,}14$).



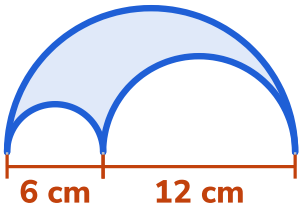
1. Yarim doira va chorak doira birga — aylananing $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ qismi.
2. Yoylar: $\frac{3}{4} \cdot 2 \cdot 3{,}14 \cdot 12 = 56{,}52$ cm.
3. Punktir chiziq qo'shilmaydi: $56{,}52 + 12 + 12 = 80{,}52$ cm.

USUL · FAQAT YOYLAR BILAN CHEGARALANGAN SHAKL

1. Har bir yoy qaysi aylananing qismi ekanini aniqlang: yarmimi yoki choragimi.
2. Har bir yoyning uzunligini toping.
3. Yoylar uzunliklarini qo'shing — bu shaklning perimetri.

1-MISOL · UCHTA YARIM AYLANA

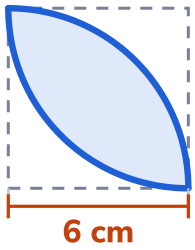
Shakl bitta katta va ikkita kichik yarim aylana bilan chegaralangan. Perimetrini toping ($\pi = 3{,}14$).



1. Kichik yoylar: $\frac{1}{2} \cdot 3{,}14 \cdot 6 = 9{,}42$ cm, $\frac{1}{2} \cdot 3{,}14 \cdot 12 = 18{,}84$ cm.
2. Katta diametr: $6 + 12 = 18$ cm. Yoyi: $\frac{1}{2} \cdot 3{,}14 \cdot 18 = 28{,}26$ cm.
3. Perimetr: $9{,}42 + 18{,}84 + 28{,}26 = 56{,}52$ cm.

2-MISOL · IKKITA CHORAK AYLANA

Kvadratning tomoni 6 cm. Uning ikki qarama-qarshi uchidan radiusi 6 cm bo'lgan chorak aylanalar chizildi. Bo'yalgan qism perimetrini toping ($\pi = 3{,}14$).



1. Har bir yoy — radiusi 6 cm li aylananing choragi: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 6 = 9{,}42$ cm.
2. Ikkita yoy: $2 \cdot 9{,}42 = 18{,}84$ cm.

2-QISM

Misollar ishlash



Shakl ichidagi punktir chiziqlar perimetrga kirmaydi. Faqat shaklni o'rab turgan chiziq hisoblanadi.

1-MISOL · YUZDAN BIRGACHA YAXLITLASH

Yarim doira shaklidagi gulzorning radiusi 13 m. Uning perimetrini toping ($\pi = 3{,}141$). Javobni yuzdan birlar xonasigacha yaxlitlang.

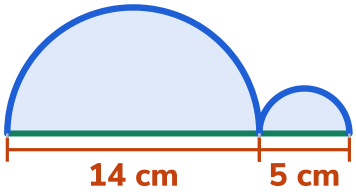


1. Diametr: $2 \cdot 13 = 26$ m.
2. Yoy: $\frac{1}{2} \cdot 3{,}141 \cdot 26 = 40{,}833$ m.
3. Perimetr: $40{,}833 + 26 = 66{,}833 \approx 66{,}83$ m.

$\pi = 3{,}14$ bo'lsa, perimetr $40{,}82 + 26 = 66{,}82$ m chiqardi — π shartdagidek olinadi. π orqali: yoy $\frac{1}{2} \cdot 26 \cdot \pi = 13\pi$, perimetr $(13\pi + 26)$ m.

2-MISOL · BITTA CHIZIQDA IKKITA YARIM DOIRA

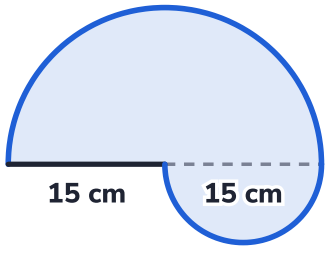
Shakl ikkita yarim doiradan iborat, diametrlari bitta to'g'ri chiziqda. Perimetrini toping ($\pi = 3{,}14$).



1. Katta yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 14 = 21{,}98$ cm. Kichik yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 5 = 7{,}85$ cm.
2. Pastdagi to'g'ri chiziq — ikkala diametr: $14 + 5 = 19$ cm.
3. Perimetr: $21{,}98 + 7{,}85 + 19 = 48{,}83$ cm.

3-MISOL · DIAMETRNING BIR QISMI SHAKL ICHIDA

Katta yarim doiraning diametri 30 cm. Diametrning o'ng yarmiga pastdan diametri 15 cm bo'lgan kichik yarim doira yopishtirildi. Shaklning perimetrini toping ($\pi = 3{,}14$).



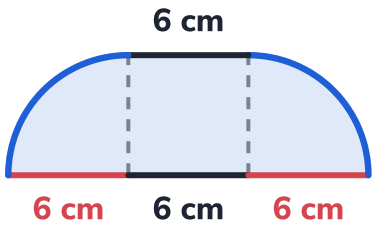
1. Katta yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 30 = 47{,}1$ cm. Kichik yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 15 = 23{,}55$ cm.
2. Katta diametrning o'ng yarmi shakl ichida qoldi (punktir). Chegarada faqat chap yarmi bor: 15 cm.
3. Perimetr: $47{,}1 + 23{,}55 + 15 = 85{,}65$ cm.

Kichik yarim doira kesib olinsa (yoyi katta yarim doira ichiga qaraydi), perimetr o'zgarmaydi: o'sha ikki yoy va 15 cm.

4-MISOL · KVADRAT VA CHORAK DOIRALAR

Shakl kvadrat va ikkita chorak doiradan iborat. Kvadratning tomoni 6 cm, chorak doiralarning radiusi ham 6 cm. Perimetrini toping ($\pi = 3{,}14$).

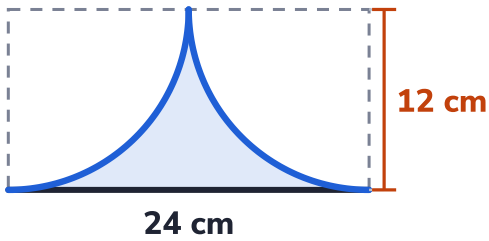
1. Ikkita yoy: $2 \cdot \frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 6 = 18{,}84$ cm.
2. To'g'ri chiziqlar: pastda $6 + 6 + 6 = 18$ cm, tepada 6 cm. Kvadratning yon tomonlari shakl ichida — ular qo'shilmaydi.
3. Perimetr: $18{,}84 + 18 + 6 = 42{,}84$ cm.

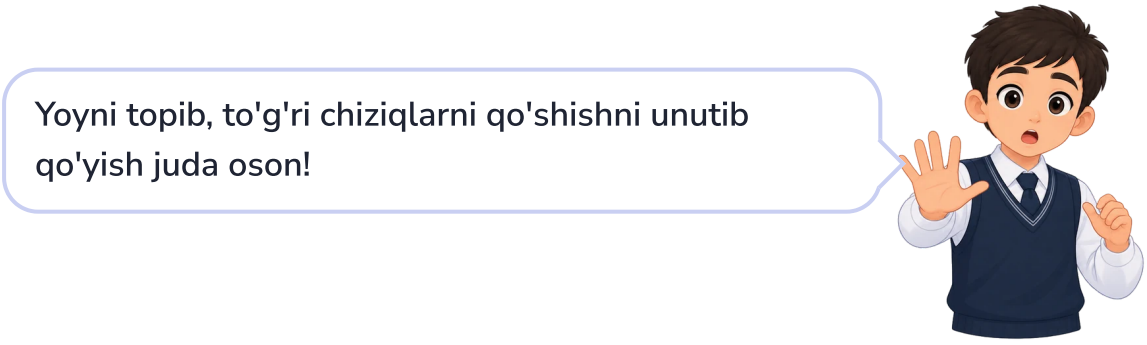


5-MISOL · ICHKARIGA BOTGAN YOYLAR

To'g'ri to'rtburchakning bo'yi 24 cm, eni 12 cm. Uning tepadagi ikki burchagidan radiusi 12 cm bo'lgan chorak doiralar kesib olindi. Qolgan shaklning perimetrini toping ($\pi = 3{,}14$).

1. Yoy ichkariga botgan bo'lsa ham, uning uzunligi o'zgarmaydi: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 12 = 18{,}84$ cm.
2. Ikkita yoy: $2 \cdot 18{,}84 = 37{,}68$ cm. To'g'ri chiziq faqat pastda: 24 cm.
3. Perimetr: $37{,}68 + 24 = 61{,}68$ cm.





DIQQAT!

1. Yarim doirada diametr ham qo'shiladi. $d = 12$ cm: ✗ 18,84 cm ✓ $18{,}84 + 12 = 30{,}84$ cm
2. Chorak doirada ikkita radius qo'shiladi. $r = 6$ cm: ✗ $9{,}42 + 6$ ✓ $9{,}42 + 6 + 6 = 21{,}42$ cm
3. Radius berilsa, yoy uchun avval diametrni toping. $r = 11$ cm: ✗ $\frac{1}{2} \cdot 3{,}14 \cdot 11$ ✓ $\frac{1}{2} \cdot 3{,}14 \cdot 22 = 34{,}54$ cm
4. Chorak doira perimetri yarim doira perimetrining yarmi emas. $d = 12$ cm: ✗ $30{,}84 : 2$ ✓ 21,42 cm
5. Shakl ichidagi punktir chiziqlar perimetrga qo'shilmaydi.

3-QISM

Mustahkamlash

1. Bo'sh joyni to'ldiring.
   a) Yarim aylana perimetri = yarim aylana yoyi uzunligi + ...
   b) Chorak aylananing perimetri = yoy uzunligi + ... + ...
   c) Yarim aylana yoyining uzunligi aylana uzunligining ... ga teng.
   d) Chorak doira yoyining uzunligi aylana uzunligining ... qismiga teng.

2. Yarim doira perimetrini toping ($\pi = 3{,}14$).
   a) $d = 20$ cm
   b) $d = 7$ m
   c) $r = 4$ cm
   d) $r = 1{,}5$ dm.

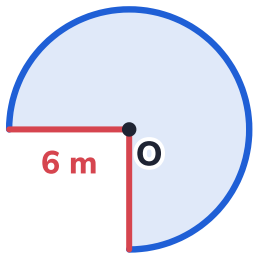
3. Yarim doira perimetrini toping ($\pi = \frac{22}{7}$).
   a) $d = 42$ cm
   b) $r = 1{,}4$ m
   c) $d = 0{,}7$ m.

4. Chorak doira perimetrini toping ($\pi = 3{,}14$).
   a) $r = 15$ cm
   b) $r = 2$ m
   c) $r = 30$ mm
   d) butun doiraning diametri 16 cm.

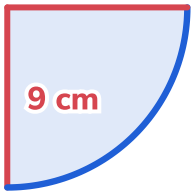
5. Chorak doira perimetrini toping ($\pi = \frac{22}{7}$).
   a) $r = 70$ cm
   b) $r = 0{,}7$ m
   c) $r = 5{,}6$ dm.

6. Perimetrni toping va javobni yuzdan birlar xonasigacha yaxlitlang.
   a) Yarim doira, $d = 9$ cm (kalkulyatordagi π tugmasi bilan).
   b) Chorak doira, $r = 7{,}5$ cm ($\pi = 3{,}141$).

7. Shakllarning perimetrini toping ($\pi = 3{,}14$).

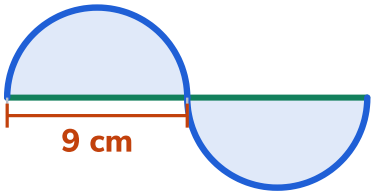


a) aylananing $\frac{3}{4}$ qismi, O — markaz

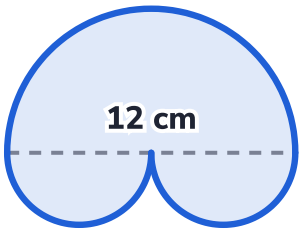


b) chorak doira

8. Shakl 2 ta bir xil yarim doiradan iborat. Uning perimetrini toping ($\pi = 3{,}14$).



9. Bo'yalgan shakl bitta katta va ikkita bir xil kichik yarim doiradan iborat. Uning perimetrini toping ($\pi = 3{,}14$).



10. Yarim doira shaklidagi gilamning diametri 1,4 m. Uning cheti bo'ylab hoshiya tikiladi. Necha metr hoshiya kerak? ($\pi = \frac{22}{7}$)

11. Bog'dagi maysazor chorak doira shaklida, radiusi 20 m. Uning atrofiga past to'siq qo'yiladi. To'siqning 1 metri 25 000 so'm. To'siq uchun necha so'm kerak? ($\pi = 3{,}14$)

4-QISM

Mustaqil ish

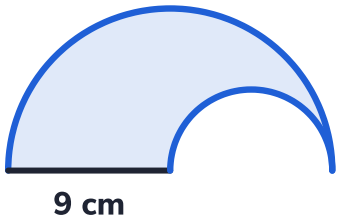
1. Diametri 30 cm bo'lgan yarim doiraning perimetrini toping ($\pi = 3{,}14$).

2. Radiusi 2,8 m bo'lgan chorak doiraning perimetrini toping ($\pi = \frac{22}{7}$).

3. Yarim doiraning radiusi 6,5 cm. Uning perimetrini kalkulyatordagi π tugmasi bilan toping va yuzdan birlar xonasigacha yaxlitlang.

4. Aylananing $\frac{3}{4}$ qismidan iborat shaklning radiusi 4,2 dm. Shaklning perimetrini toping ($\pi = \frac{22}{7}$).

5. Katta yarim doiraning diametri 18 cm. Undan diametri 9 cm bo'lgan kichik yarim doira kesib olindi. Qolgan shaklning perimetrini toping ($\pi = 3{,}14$).



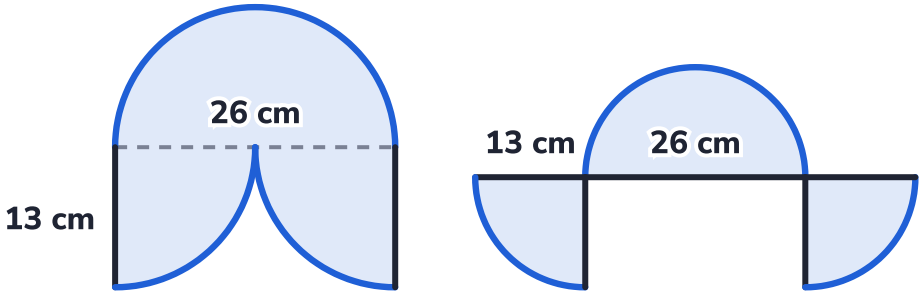
6. Shakl tomoni 4 cm bo'lgan kvadrat va chorak doiradan iborat. Perimetrini toping ($\pi = 3{,}14$).



7. Yarim doira shaklidagi sahnaning diametri 9 m. Uning cheti bo'ylab chiroqli tasma tortiladi. 25 m tasma yetadimi? Yetsa, qancha ortib qoladi? ($\pi = 3{,}14$)

8. **Xatoni toping.** Bekzod radiusi 16 cm bo'lgan chorak doiraning perimetrini shunday topdi: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 16 + 16 = 41{,}12$ cm. Xato nimada? To'g'ri javob qancha?

9. **Qiyinroq.** Diametri 26 cm bo'lgan yarim doira va radiusi 13 cm bo'lgan ikkita bir xil chorak doiradan ikki xil shakl tuzildi ($\pi = 3{,}14$).
   a) 1-shaklning perimetrini toping.
   b) 2-shaklning perimetrini toping. U 1-shaklnikidan qancha uzun?



1-shakl 2-shakl

5-QISM

Javoblar

Mustahkamlash

1-masala

a) diametr; b) radius + radius; c) yarmiga; d) $\frac{1}{4}$ qismiga (choragiga).

2-masala

a) $\frac{1}{2} \cdot 3{,}14 \cdot 20 = 31{,}4$; $31{,}4 + 20 = 51{,}4$ cm; b) $\frac{1}{2} \cdot 3{,}14 \cdot 7 = 10{,}99$; $10{,}99 + 7 = 17{,}99$ m; c) $d = 8$ cm, $12{,}56 + 8 = 20{,}56$ cm; d) $d = 3$ dm, $4{,}71 + 3 = 7{,}71$ dm.

3-masala

a) $\frac{1}{2} \cdot \frac{22}{7} \cdot 42 = 66$; $66 + 42 = 108$ cm; b) $d = 2{,}8$ m, $\frac{1}{2} \cdot \frac{22}{7} \cdot 2{,}8 = 4{,}4$; $4{,}4 + 2{,}8 = 7{,}2$ m; c) $\frac{1}{2} \cdot \frac{22}{7} \cdot 0{,}7 = 1{,}1$; $1{,}1 + 0{,}7 = 1{,}8$ m.

4-masala

a) $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 15 = 23{,}55$; $23{,}55 + 15 + 15 = 53{,}55$ cm; b) $3{,}14 + 2 + 2 = 7{,}14$ m; c) $47{,}1 + 30 + 30 = 107{,}1$ mm; d) $r = 16 : 2 = 8$ cm, $12{,}56 + 8 + 8 = 28{,}56$ cm.

5-masala

a) $\frac{1}{4} \cdot 2 \cdot \frac{22}{7} \cdot 70 = 110$; $110 + 70 + 70 = 250$ cm; b) $1{,}1 + 0{,}7 + 0{,}7 = 2{,}5$ m; c) $8{,}8 + 5{,}6 + 5{,}6 = 20$ dm.

6-masala

a) Yoy $\approx 14{,}137$ cm; $14{,}137 + 9 = 23{,}137 \approx 23{,}14$ cm. b) Yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}141 \cdot 7{,}5 = 11{,}77875$; $11{,}77875 + 15 = 26{,}77875 \approx 26{,}78$ cm.

7-masala

a) $\frac{3}{4} \cdot 2 \cdot 3{,}14 \cdot 6 = 28{,}26$; $28{,}26 + 6 + 6 = 40{,}26$ m. b) $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 9 = 14{,}13$; $14{,}13 + 9 + 9 = 32{,}13$ cm.

8-masala

Har bir yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13$ cm. Ikkala diametr ham chegarada: $14{,}13 + 14{,}13 + 9 + 9 = 46{,}26$ cm.

9-masala

Katta yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 12 = 18{,}84$ cm. Kichik yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 6 = 9{,}42$ cm. Punktir diametr qo'shilmaydi: $18{,}84 + 9{,}42 + 9{,}42 = 37{,}68$ cm.

10-masala

$\frac{1}{2} \cdot \frac{22}{7} \cdot 1{,}4 = 2{,}2$ m; $2{,}2 + 1{,}4 = 3{,}6$ m.

11-masala

Yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 20 = 31{,}4$ m. Perimetr: $31{,}4 + 20 + 20 = 71{,}4$ m. Narxi: $71{,}4 \cdot 25\,000 = 1\,785\,000$ so'm.

Mustaqil ish

1-masala

$\frac{1}{2} \cdot 3{,}14 \cdot 30 = 47{,}1$; $47{,}1 + 30 = 77{,}1$ cm.

2-masala

$\frac{1}{4} \cdot 2 \cdot \frac{22}{7} \cdot 2{,}8 = 4{,}4$; $4{,}4 + 2{,}8 + 2{,}8 = 10$ m.

3-masala

$d = 13$ cm. Yoy $\approx 20{,}420$ cm; $20{,}420 + 13 = 33{,}420 \approx 33{,}42$ cm.

4-masala

$\frac{3}{4} \cdot 2 \cdot \frac{22}{7} \cdot 4{,}2 = 19{,}8$; $19{,}8 + 4{,}2 + 4{,}2 = 28{,}2$ dm.

5-masala

Katta yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 18 = 28{,}26$ cm. Kichik yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13$ cm. Perimetr: $28{,}26 + 14{,}13 + 9 = 51{,}39$ cm.

6-masala

Yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 4 = 6{,}28$ cm. Pastda $4 + 4 = 8$ cm, tepada 4 cm, chapda 4 cm. Perimetr: $6{,}28 + 8 + 4 + 4 = 22{,}28$ cm.

7-masala

$\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13$; $14{,}13 + 9 = 23{,}13$ m. Yetadi, ortib qoladi: $25 - 23{,}13 = 1{,}87$ m.

8-masala

Bekzod bitta radiusni qo'shmadi. To'g'risi: $25{,}12 + 16 + 16 = 57{,}12$ cm.

9-masala

Yarim doira yoyi: $\frac{1}{2} \cdot 3{,}14 \cdot 26 = 40{,}82$ cm. Chorak doira yoyi: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 13 = 20{,}41$ cm. a) Diametr shakl ichida (punktir): $40{,}82 + 20{,}41 + 20{,}41 + 13 + 13 = 107{,}64$ cm. b) Bo'laklar faqat uchlari bilan tegib turibdi, hamma chiziq chegarada: $40{,}82 + 26 + 2 \cdot (13 + 20{,}41 + 13) = 159{,}64$ cm; $159{,}64 - 107{,}64 = 52$ cm uzun (diametr 26 cm va ikkita radius 13 + 13 cm).